

# L2 Booting the Project Phase 1



This lab focuses on booting up the project phase I.

We will first present and discuss the sample source code provided (see drive).

After that you may start to develop the step #0, related to implementing the **R0** fragment of the Rusty language, as specified in the next slide.

# Towards the Rusty language (step #0)

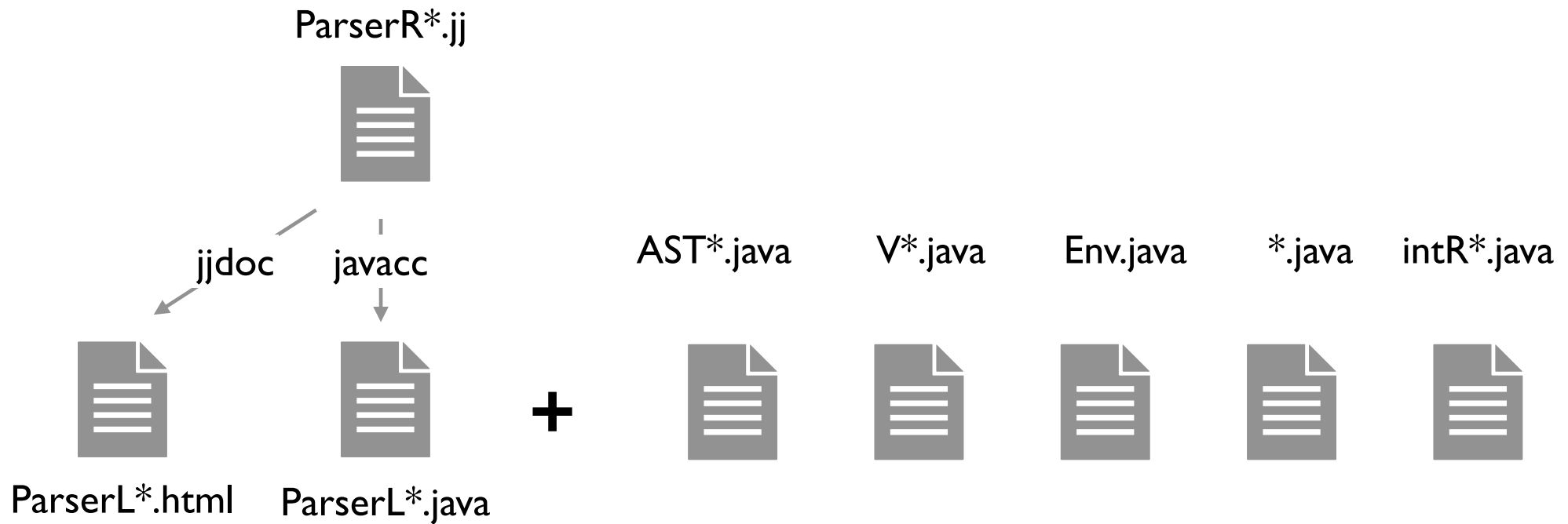
```
EE -> | num | true | false
      | EE + EE | EE - EE
      | EE * EE | EE / EE | -EE | ( EE )
      | EE == EE | EE > EE | EE >= EE | ...
      | EE && EE | EE || EE | !EE
      | let id = EE ; EE
      | { EE }
```

```
{
    let y = 1;
    let b = (y>0) && (y<=20);
    {
        let y = 2*y;
        let z = 4+y;
        println(b || !(z < 0))
    }
}
```

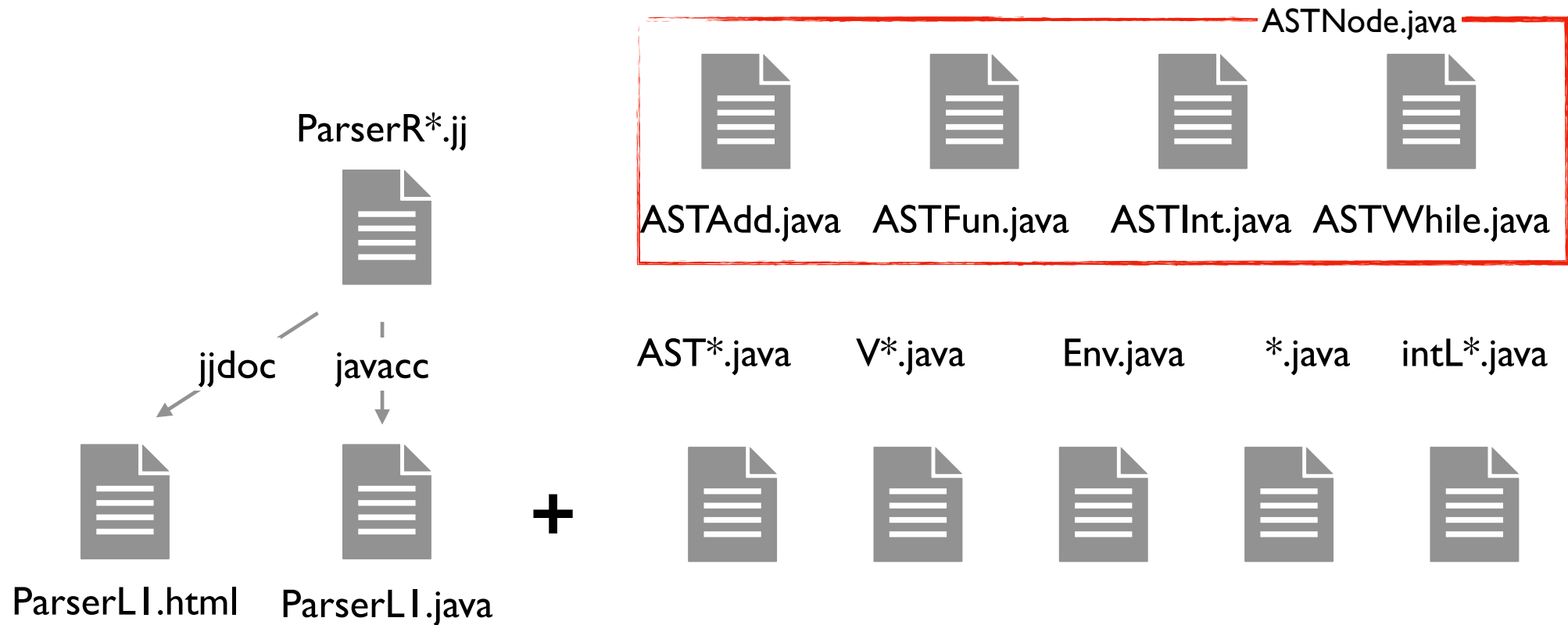
- Consider the abstract syntax above. Complete the ASTNode classes for these basic constructs, expressing basic arithmetic and logic operations and scoped **declaration blocks**.
- We also need to complete the parser and the implementation of the Environment class.
- We may also see above (on the right) an example of a small Rusty (step 0) program.

# Structure of the L\* Interpreters (overview)

# Structure of the R\* Interpreters (overview)



# Structure of the R\* Interpreter



# Structure of the R\* Interpreter

